

VISION-GROUNDED RL IN MEAL RECOMMENDATION

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1 ABSTRACT

Personalized meal recommendation systems often fail because they rely on tedious manual dietary tracking and struggle to balance competing nutritional and preference goals. To solve this, we propose an end-to-end framework that eliminates manual logging by linking a visual perception module directly to a Reinforcement Learning (RL) recommendation agent. Our core idea relies on processing meal images to jointly predict dish-level nutrition and semantic ingredient styles, which our RL agent then uses alongside user meal history to optimize multi-objective dietary suggestions. Evaluated on the Nutrition5k dataset, our OpenCLIP-based perception module achieves 21.6 (g) MAE for nutrition regression and 0.59 micro-F1 / 0.63 micro-mAP for multi-label ingredient classification, improving over the baseline by 37.6% in MAE, 45.6% in micro-F1, and 77.6% in micro-mAP. Evaluated against proposed baselines, our meal suggestion agent improves nutritional goal attainment by $\sim 20\%$ points at the 21-day horizon while maintaining preference alignment and dietary diversity. The return advantage over the myopic multi-objective baseline grows with the planning horizon, suggesting that learned temporal credit assignment is useful for long-horizon meal planning.

2 INTRODUCTION

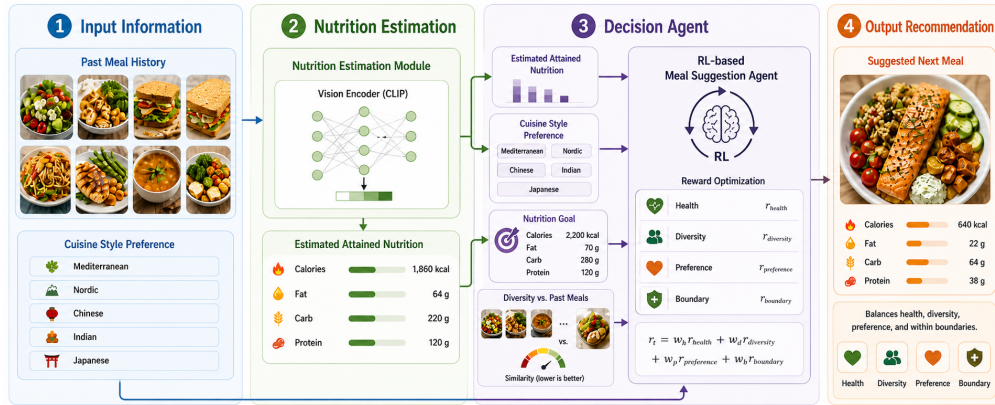


Figure 1: Overall framework of the proposed system.

Existing nutrition tracking applications typically suffer from high user friction and low accuracy due to their reliance on manual data entry. Furthermore, effective meal planning requires solving a complex multi-objective optimization problem: a sustainable diet must balance strict health targets with dietary diversity and personal taste constraints. This raises a critical research question: Can an end-to-end ML framework be used to alleviate user tracking friction while simultaneously optimizing long-term dietary adherence? To address these challenges, we propose an end-to-end framework (illustrated in Figure 1) that integrates a Computer Vision model for automated nutrition estimation with a RL-based agent for personalized meal recommendation. We aim to benefit everyday users and digital health applications by entirely removing manual logging. If successful, the implications of this work include significantly lower user friction, better generalization to long-term dietary adherence, and a more reliable pipeline for automated nutritional interventions. Our success criteria are then twofold: achieving high accuracies in visual nutrition estimation while demonstrating that our RL agent can formulate multi-day meal plans that yield higher cumulative rewards for nutritional balance and preference alignment.

3 PROBLEM FORMULATION AND RELATED WORK

Our framework builds upon and bridges three distinct lines of research: visual nutrition estimation, dietary recommendation systems, and sequential decision-making via RL. In visual nutrition estimation, datasets like Nutrition5k (Thames et al., 2021) have enabled multi-task models such as Nutriconv (Junquera et al., 2025) to predict mass and macronutrients from images. Recent work has explored using CLIP for food analysis in zero-shot settings. For example, Gatti et al. (2025) have shown that CLIP can help recognize food items from text prompts with high precision, highlighting the usefulness of vision-language models for food recognition. Traditional dietary recommendation systems primarily rely on hybrid or knowledge-based filtering (Abhari et al., 2019). However, these approaches generally optimize for single-interaction engagement rather than long-term adherence. Finally, while RL has been applied to healthcare and planning domains (Yu et al., 2021), its application to multimodal, closed-loop meal planning remains underexplored.

Unlike traditional recommendation systems that optimize for static, single-point engagement, or vision-only applications that strictly estimate nutrition without proactive planning, our approach is uniquely positioned at the intersection of both. Table 1 highlights how our end-to-end framework eliminates the manual logging bottleneck while simultaneously treating meal planning as a sequential, multi-objective optimization problem.

Paradigm	Automated Input	Sequential Planning	Multi-Objective
Traditional RecSys (Abhari et al., 2019)	–	–	Partial
Vision-Only Trackers (Junquera et al., 2025)	✓	–	–
Ours (End-to-End)	✓	✓	✓

Table 1: Positioning of our framework against existing nutritional tracking and recommendation paradigms.

We formulate personalized meal planning as a sequential recommendation problem. At each meal time step t , the system receives three inputs: a user preference representation $\mathbf{p} \in \mathbb{R}^d$, a recent meal history $\mathcal{H}_t = \{\mathbf{v}_{t-K}, \dots, \mathbf{v}_{t-1}\}$ represented by visual meal embeddings, and a target nutrition vector $\mathbf{n}_{\text{target}} \in \mathbb{R}^k$ specifying caloric and macronutrient goals. The goal is to recommend a meal a_t with visual embedding $\mathbf{v}_{a_t}$ and nutrition vector $\mathbf{n}_{a_t}$.

The state s_t summarizes the user’s remaining nutritional needs, preference embedding, and recent meal history. The desired output is a sequence of recommended meals $\{a_1, a_2, \dots, a_T\}$ over the planning horizon that balances nutrition, preference alignment, and dietary variety. We optimize a multi-objective reward:

$$R(s_t, a_t) = \alpha r_{\text{nutrition}}(s_t, a_t) + \beta r_{\text{preference}}(s_t, a_t) + \lambda r_{\text{diversity}}(s_t, a_t), \quad (1)$$

where $r_{\text{nutrition}}$ rewards reducing the gap between consumed and target nutrition, $r_{\text{preference}}$ rewards similarity between the recommended meal and the user preference embedding, and $r_{\text{diversity}}$ discourages repeatedly recommending visually similar meals. Similarity is computed using cosine similarity between normalized meal embeddings.

4 METHODOLOGY

Our framework (Figure. 1) couples a CV module with an RL policy. This approach grounds decisions in the user’s estimated intake while optimizing multi-objective goals over a planning horizon (e.g. 1, 3, 7, and 21 days in our experiments), including nutrition target attainment, dietary diversity, and preference alignment. The system operates as a closed loop: each newly recommended meal is ingested as updated state, allowing the policy to propose future recommendations.

4.1 NUTRITION ESTIMATION

Supervision Approach. The Nutrition5k dataset contains nutrition and ingredient metadata well-suited for regression and classification tasks. For regression, the dataset provides five dish-level

targets: total calories, total mass, total fat, total carbohydrate, and total protein. For ingredient classification, we filter out visually-negligible ingredients (defined as those with mass $\geq 5\text{g}$ and $\geq 2\%$ of the total dish mass). The final classification space uses the top-100 most frequent filtered ingredients in the training set, i.e. each dish is represented by a 100-dimensional multi-hot vector.

Network Architecture. Our current model uses OpenCLIP with a ViT-B-32 visual encoder pre-trained on laion2b_s34b_b79k to process Nutrition5k meal images. For each dish, we encode the available images and mean-pool the per-image embeddings to obtain a single dish-level representation. We evaluate two settings. In the baseline setting, the OpenCLIP encoder is kept frozen and the dish embedding is used to train separate downstream heads: a nutrition regressor for the five continuous targets and a multi-label ingredient classifier. In the multitask setting, a shared OpenCLIP encoder feeds two task-specific heads, one for regression and one for ingredient classification, and the model is trained jointly with $\mathcal{L}_{total} = \lambda_{reg}\mathcal{L}_{reg} + \lambda_{cls}\mathcal{L}_{cls}$. Unlike the baseline, the multi-task model is not fully frozen: after an initial freeze period, we fine-tune the last n visual transformer blocks of the encoder together with the two task heads.

Training Details. For the multitask setting, we train for 20 epochs with batch size 32, head learning rate 10^{-3} , encoder learning rate 10^{-5} , and equal multitask loss weights $\lambda_{reg} = \lambda_{cls} = 1$. The encoder is frozen for the first 3 epochs, after which we fine-tune the last 2 visual transformer blocks.

Hypothesis 1: Multi-task learning enables a better meal representation. Our regression and classification tasks rely on overlapping visual cues. If both forms of supervision are used together, the encoder may learn features that are simultaneously more discriminative and more nutritionally informative. To test this, we will compare (1) training the model using only the regression objective and evaluate its regression performance; (2) training the model using only the classification objective and evaluate the classification performance. (3) training a shared encoder with both regression and classification losses using a multi-task objective function outlined in Section. We will then compare all three settings under the same evaluation protocol.

Hypothesis 2: Training-time data augmentation improves generalization and robustness for visual nutrition estimation The Nutrition5k dataset contains a diverse set of meals over which the ingredient classifier and nutrition regression must be trained. However, because the source of information is ultimately a single frame of visual information, we must attend carefully to the features of the training subset to ensure proper generalization. We hypothesize that applying image augmentations during training will expand the effective training set and improve the model’s robustness to small appearance changes. To test this, we will conduct an ablation study comparing the performance of both our baseline and multi-task models with and without data augmentation enabled on the Nutrition5k dataset.

Evaluation procedure. We use the official Nutrition5k RGB train/test split, with a validation split carved from training data. For regression, we report MAE and RMSE. For the multi-label classification, we report micro and macro mean average precision (mAP). Since the ingredient labels in Nutrition5k are highly long-tail imbalanced, micro-averaged metrics help to measure overall performance dominated by frequent classes, and macro-averaged metrics help to assess whether the model also performs well on rare ingredients. In addition to mAP, we also report micro and macro F1 score. Finally, we include Precision@5 and Recall@5 to further evaluate the model’s top-5 ingredient prediction for each dish.

4.2 MEAL SUGGESTION AGENT

We formulate meal recommendation as a finite-horizon Markov Decision Process (MDP). Each episode spans $H = N_{\text{days}} \times M_{\text{meals}}$ meal decisions, where N_{days} is the planning horizon and $M_{\text{meals}} = 3$ corresponds to breakfast, lunch, and dinner. At each step, the agent recommends one meal template and one portion size from the catalog. The objective is to maximize long-term reward without relying on expert demonstrations or ground-truth meal plans.

State. The state s_t summarizes the user’s nutritional status, temporal context, recent meal history, and preferences. Specifically, the observations include (1) the normalized remaining daily nutrition deficit; (2) the normalized remaining episode-level nutrition deficit; (3) the daily nutrition target; (4) the remaining fraction of the episode; (5) the mean-pooled embedding of the most recent K consumed meals; and (6) the user’s preference embedding. Nutrition vectors contain calories, protein,

carbohydrates, and fat. Meal and preference embeddings lie in the same 512-dimensional CLIP embedding space.

Action. The action space is discrete. Each action a_t decodes into a meal template m_t and portion multiplier q_t : $a_t = (m_t, q_t)$. Specifically, the action space contains 1,000 meal templates from a preset catalog and 3 portion levels, giving 3,000 possible meal-portion actions. The selected action determines both a visual embedding $\mathbf{e}_{a_t}$ and a portion-scaled nutrition vector $\mathbf{n}_{a_t}$.

Reward. The reward combines nutrition progress, dietary diversity, preference alignment, and boundary adherence:

$$r_t = w_h r_{\text{health}} + w_d r_{\text{diversity}} + w_p r_{\text{preference}} + w_b r_{\text{boundary}}.$$

r_{health} rewards reduction in the mean absolute nutrition deficit after normalizing each nutrient by its daily target. This prevents calories from dominating smaller-scale macronutrient targets. $r_{\text{diversity}}$ encourages the agent to avoid recommending meals visually similar to recent history:

$$r_{\text{diversity}} = 1 - \cos(\mathbf{e}_{a_t}, \bar{\mathbf{h}}_t),$$

where $\bar{\mathbf{h}}_t$ is the normalized mean embedding of recent meals. $r_{\text{preference}}$ rewards alignment with the user’s preference embedding:

$$r_{\text{preference}} = \cos(\mathbf{e}_{a_t}, \mathbf{p}_{\text{user}}).$$

r_{boundary} is applied at the end of each day to reward closing the daily nutrition deficit. At the final episode step, an additional terminal bonus rewards overall episode-level deficit closure across the full planning horizon.

Action-Scoring Q-Network. We use an action-scoring DQN architecture to better exploit structure in the large discrete action space. Instead of treating each meal-portion pair as an individual action, each action is represented by catalog features consisting of the meal embedding, scaled nutrition vector, and portion multiplier. The state encoder first maps the observation into a latent state representation, while a separate action encoder maps each candidate action feature vector into the same latent space. For a state s_t and candidate action a , the network computes a compatibility score between the state embedding and the action embedding. Both embeddings are L2-normalized and scored with a dot product, giving a cosine-style action score. A learned action bias and state-value branch are added to produce the final Q-value:

$$Q_\theta(s_t, a) = V_\theta(s_t) + A_\theta(s_t, a),$$

where $V_\theta(s_t)$ is the state-value term and $A_\theta(s_t, a)$ is the action-specific advantage score. The advantage is centered across actions, following the intuition of dueling Q-networks. This design allows the policy to share information across visually and nutritionally similar meals: two actions with similar embeddings and nutrition profiles receive related action representations, so learning about one meal-portion pair can improve value estimates for similar candidates. This is especially useful for a catalog with thousands of discrete actions.

Hypothesis 3: DQN’s advantage over myopic multi-objective baseline grows with the planning horizon. For a single-day horizon, a greedy agent that maximizes the same composite objective without lookahead can approximate the optimal policy reasonably well, because credit assignment spans only 3 steps. As the horizon extends to 3, 7, or longer days, however, decisions at breakfast must account for how they constrain achievable diversity and deficit reduction across subsequent meals and days. We hypothesize our action-scoring DQN’s learned value function internalizes these temporal dependencies, whereas a greedy agent has no mechanism to reason across day boundaries. We test this in Section 6.2.

Training pipeline and hyperparameters The RL training pipeline consists of four stages. (1) the construction of meal catalog, detailed in 5.2. (2) initialize simulated users by sampling nutrition targets and generating preference embeddings from style-matched recipe templates; We also bootstrap recent meal history using Nutrition5K-derived meal embeddings and nutrition values, simulating a user who has already consumed prior meals. (3) train the agent in the meal-planning environment.

During training, nutrition targets are sampled independently from predefined adult target ranges:

$$\text{calories} \in [1500, 3200], \quad \text{protein} \in [50, 200], \quad \text{carbs} \in [130, 450], \quad \text{fat} \in [35, 100].$$

Preference embeddings are generated by sampling meal templates from preset dietary styles and mean-pooling their normalized CLIP embeddings. The agent is implemented using Gymnasium (Towers et al., 2024) and Stable-Baselines3 DQN (Raffin et al., 2021). We use the following default hyperparameters: learning rate 1×10^{-4} , replay buffer size 100,000, batch size 128, discount factor $\gamma = 0.99$, target-network update interval 500, exploration fraction 0.3, initial $\epsilon = 1.0$, final $\epsilon = 0.05$, and 1,000,000 total training steps. The reward weights are $w_h = 3.0$, $w_d = 0.1$, $w_p = 0.1$, and $w_b = 1.0$. The history length is $K = 6$, and the portion multipliers are $\{0.75, 1.0, 1.25\}$.

The RL agent is trained for a fixed number of steps and the final checkpoint is selected for evaluations. Agent training/evaluations are conducted on an Macbook Air M4.

Evaluation Strategy We evaluate the RL agent on held-out simulated users and dietary styles. The evaluation pool is constructed as a grid of 5 nutrition personas, 8 held-out dietary styles, and 2 random seeds, giving 80 evaluation episodes. The held-out dietary styles are disjoint from those used in training, allowing us to test whether the policy generalizes to unseen preference distributions. To test the horizon-scaling hypothesis in Section 4.2, we evaluate planning horizons $N_{\text{days}} \in \{1, 3, 7, 21\}$ while holding other hyperparameters fixed. We compare our action-scoring DQN against three baselines. (1) random policy: samples uniformly from the action space. (2) health-only greedy baseline: chooses the meal-portion pair that most reduces the current daily nutrition deficit. (3) multi-objective greedy baseline, which scores each action using the same health, diversity, and preference terms as the DQN reward, but performs only one-step lookahead and does not learn a value function.

We report four metrics. (1). **Nutrition Goal Attainment (NGA)** measures the fraction of days where the final daily deficit is within 10% of the total daily target (2). **Dietary Diversity Score (DDS)** measures variety among selected meals:

$$\text{DDS} = 1 - \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \cos(\mathbf{e}_i, \mathbf{e}_j),$$

where $\mathcal{P}$ is the set of all unordered meal pairs in the episode. (3). **Preference Alignment (PA)** is the mean cosine similarity between selected meal embeddings and the user’s preference embedding. (4). **Episodic Deficit Closure (EDC)** measures how much of the episode-level nutrition target is satisfied by the end of the planning horizon:

$$\text{EDC} = 1 - \frac{\|\Delta_{\text{episode, end}}\|_1}{\|\mathbf{n}_{\text{target}} \times N_{\text{days}}\|_1}.$$

Together, these metrics evaluate whether the agent balances nutritional correctness, user preference, dietary variety, and long-horizon planning.

5 DATA

Our project utilizes two main datasets, Nutrition5k (Thames et al., 2021) for multi-task CLIP fine-tuning, and DataHiveAI’s Recipes with Nutrition dataset (DataHiveAI, 2025) for the creation of a diverse catalog of meal suggestions.

5.1 NUTRITION5K

Preprocessing and Data Splits. We use the standard OpenCLIP image preprocessing pipeline for all images. When training augmentation is enabled, we apply augmentation only to the training split with two image variants: a horizontal flip and a mild color jitter, in addition to the original image. This expands the effective training set and improves robustness to small appearance changes. We keep the official Nutrition5k test split. The official training split is further divided with a seeded random split into training and validation subsets; in the current setup, 20% of the official training data is used for validation and the remaining 80% for training.

5.2 MEAL CATALOG

The Recipes with Nutrition dataset is used to construct a catalog of 1,000 meal options which share the same underlying ingredients as the Nutrition5k dataset, while selecting for a diversity of ingredients and cuisines. This ensures that selected recipes both transfer cleanly into the fine-tuned CLIP space while providing for users’ individual preferences and nutritional needs.

DataHiveAI Recipes Dataset (DataHiveAI, 2025) is a tabular dataset containing 39,447 recipes sourced from publicly available recipe websites. Each recipe consists of: basic recipe information like name, source, and url; structured categorical labels like meal_type, cuisine_type, diet_labels, health_labels, etc; and ingredient and nutrition information like ingredients, total_nutrients, digest, etc. Importantly, each recipe comes with an *image_url* field, which links to a photograph of the food. This is used to embed each selected recipe entry. One major limitation is that they are often natural food photography, in contrast to the strict overhead images contained in the Nutrition5k dataset. The recipes are heavily biased in cuisine towards American dishes, which make up 17,000 of the 40,000 recipes. The next largest class is Italian with 4,480 recipes, while the smallest two classes have 84 and 44 recipes respectively. We account for this in the final catalog by enforcing quotas of various styles to ensure that we have a diverse selection of cuisines.

Meal Catalog Construction In order to select the 1,000-recipe catalog, we first filter by excluding recipes with ingredients outside the Nutrition5k dataset. We also filter by meal_type to exclude desserts, drinks, and seasoning mixes and deduplicate by image_urls to exclude recipes where the food photograph is a company logo. The final set of 1,000 recipes is chosen from this filtered subset¹ using a score based on the average weighted rarity of the ingredients. The score is calculated as follows:

$$\text{score}(r) = \frac{1}{|I_r|} \sum_{i \in I_r} \frac{\log(1 + f(i))}{1 + \gamma c(i)}$$

where I_r is the set of ingredients in the recipe, $f(i)$ is the number of occurrences of an ingredient in the filtered set of recipes and $c(i)$ is the number of occurrences of an ingredient in the currently selected set of recipes, γ is a decay factor used to decide how much to penalize common ingredients. This scoring encourages meals which utilize common ingredients, while the decay factor ensures that the catalog also contains diverse ingredients. Additionally, each cuisine type is given a quota to ensure it gets an equal share of the final catalog. If there are not enough recipes of a given cuisine, the remaining quota is distributed evenly to the remaining cuisines.² At each step, the highest-scoring recipe is chosen and scoring for affected recipes are updated. If the quota of a given cuisine is met, its remaining recipes are removed from the pool. This final set of recipes is embedded in the same $\mathbb{R}^{512}$ CLIP embedding space.

6 EXPERIMENTS AND RESULTS

6.1 NUTRITION ESTIMATION

Figure 2 summarizes the 2×2 ablation over model type (baseline and multitask) and data augmentation (off and on) on Nutrition5k overhead RGB. The strongest result is obtained by the multitask model with data augmentation at training time, which achieves an overall nutrition MAE of 21.6, compared with 34.6 for the baseline without augmentation. On ingredient prediction, the same model improves micro/macro mAP from 0.353/0.266 to 0.625/0.536 and micro/macro F1 from 0.406/0.171 to 0.587/0.485, while also considerably raising Precision@5 from 0.300 to 0.379 and Recall@5 from 0.588 to 0.734.

Two trends are clear from the ablation. First, multitask fine-tuning is the dominant source of improvement: even without data augmentation, the multitask model substantially outperforms the augmented baseline on both regression and classification metrics. Second, data augmentation provides consistent but smaller gains within each model family, indicating that robustness to appearance variation helps but does not account for most of the performance gain. Overall, these results suggest that jointly adapting the visual encoder for nutrition regression and ingredient recognition produces a more informative dish representation than frozen CLIP embeddings with separate linear probes. Our experiments directly test Hypotheses 1 and 2: joint regression-classification training yields a better shared representation, and that training-time augmentation improves generalization.

¹8,391 filtered recipes

²The cuisine type/style used in RL Training: American, Asian, Mediterranean, Central European, Nordic, Chinese, Indian, Japanese; for evaluation: French, Italian, Mexican, South American, Eastern European, British, Middle Eastern, South East Asian

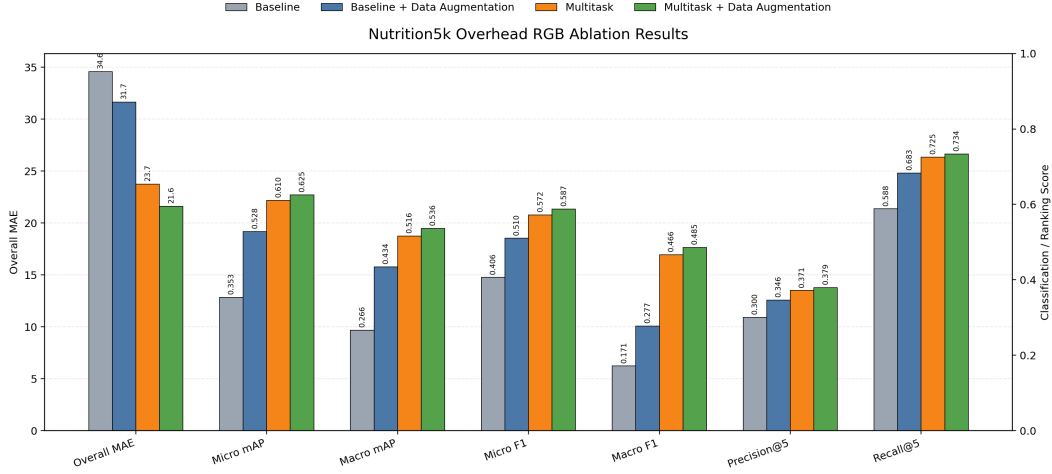


Figure 2: Ablation results on Nutrition5k overhead RGB for baseline vs. multitask training, with and without training augmentation.

6.2 ACTION-SCORING DQN VS. GREEDY BASELINES

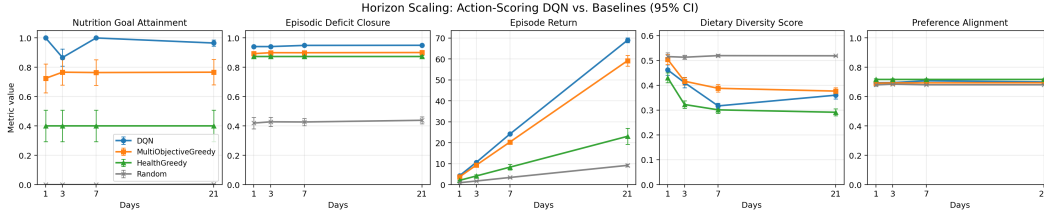


Figure 3: Horizon-scaling performance of the action-scoring DQN and baseline policies. Points show the mean over 80 held-out evaluation episodes, and error bars show 95% confidence intervals.

We evaluate whether the action-scoring DQN benefits more from temporal credit assignment as the planning horizon increases. We compare against three baselines: a random policy, a health-only greedy policy, and a multi-objective greedy policy that uses the same immediate reward terms as DQN but performs no lookahead.

The result is shown in 3. Across horizons, DQN achieves the highest NGA, EDC, and return. At 7 days, DQN reaches 1.000 NGA and 0.949 EDC, compared with 0.764 NGA and 0.899 EDC for the multi-objective greedy baseline. At 21 days, DQN reaches 0.965 NGA and 0.950 EDC, outperforming the multi-objective greedy baseline at 0.767 NGA and 0.901 EDC. These results support our hypothesis 3: learning a value function helps the agent coordinate nutrition decisions across meal slots and longer planning windows. The random policy obtains the highest DDS because it samples meals broadly, but it performs poorly on nutrition metrics, showing that diversity alone is not sufficient for meal planning. The greedy baselines are competitive but limited by their myopic decision rules. HealthGreedy minimizes the immediate remaining daily nutrition deficit at each meal, but it does not plan how the remaining meal slots should collectively satisfy the daily target; as a result, early choices can overshoot some macros while leaving others difficult to repair. MultiObjective-Greedy improves the one-step tradeoff among health, diversity, and preference, but it still lacks a value estimate for future consequences. In contrast, DQN can learn action values that account for downstream meal choices and threshold-based daily adherence.

Long-horizon agent improvement. Our initial long-horizon runs showed weaker 21-day performance, indicating that temporal credit assignment becomes harder as the episode length grows. We therefore introduced current time-aware state representation and a lower discount factor, $\gamma = 0.95$, while keeping the action-scoring architecture and reward weights fixed. This improved 21-day DQN

performance substantially, suggesting that long-horizon meal planning benefits from explicitly encoding temporal progress and reducing overemphasis on very distant rewards.



Figure 4: UMAP visualization of catalog embedding

Embedding-space failure mode. The catalog UMAP visualization shown in 4 reveals that recipe styles are not cleanly clustered in CLIP embedding space. Points from different cuisine/style labels are heavily intermingled, suggesting that CLIP primarily organizes meals by visual appearance rather than dietary style. This weakens the assumption that style-based preference embeddings will naturally retrieve same-style meals. A future fix is to add style- or ingredient-aware representation learning, such as supervised contrastive fine-tuning, metadata-conditioned embeddings, or preference embeddings learned from user feedback rather than style labels alone.

7 CONCLUSION

In conclusion, our framework integrates CV model for automated nutrition estimation with a RL agent to optimize long-term meal planning. Our evaluation shows that multitask fine-tuning, combined with data augmentation, proved effective. This joint training approach achieved an overall nutrition MAE of 21.6 g and improved multi-label ingredient classification. Furthermore, our Action-Scoring DQN successfully maximized long-term dietary goals, outperforming random and myopic greedy baselines. The agent achieved a NGA of 1.00 at a 7-day planning horizon and 0.965 at a 21-day horizon.

Limitations exist. First, fine-tuning the visual encoder for nutrition estimation risks shifting the pre-trained text-image-aligned embedding space. Furthermore, UMAP visualizations revealed that CLIP organizes meals primarily by visual appearance rather than dietary style. Our assumption that style-based preference embeddings will reliably retrieve same-style meals was weakened. Second, the DataHiveAI catalog dataset contains a heavy bias toward American cuisine and consists of natural food photography, which contrasts with the strictly overhead images found in the Nutrition5k training dataset. Third, estimating portion size from single-view RGB image is inherently ambiguous. Prior work (Nogay et al., 2025) has reframed portion estimation as a coarse classification over discrete weight intervals rather than gram-level regression.

Informed by our results and the limitations, our next steps are (1). addressing embedding drift, future iterations should re-anchor the fine-tuned encoder back to the shared vision-language space using contrastive learning applied to recipe text-image pairs. To fix the embedding-space clustering issue, we consider adding style-aware representation learning. This could involve supervised contrastive fine-tuning, or deriving preference embeddings directly from user feedback rather than static style labels. (2). deriving more robust portion estimation. One viable method is to transition to video-based dietary assessment. By capturing a short video sequence of the meal, the system can leverage multi-view shape recovery methods to reconstruct a 3D geometric model of the food, which allows for a more accurate calculation of food volume.

8 TEAM CONTRIBUTION

Team Member	Contribution
Tsung-Huan Yang	Implementation, experiments, analysis of RL agent. Participation in writing section 1, 2, 3, 4.2, 6.2, 7
Jason Yu	Nutrition estimation implementation and experiments; participation in writing Section 3, 4.1, 5.1, and 6.1
Tiger Peng	Meal catalog construction; Participation in writing Section 3, 5, 5.2

Table 2: Team contribution table

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